Unsupervised Concept Discovery for Medical Vision–Language Models: A Rigorous Characterization of Sparse-Autoencoder Failure and Deterministic Alternatives

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ABSTRACT

Medical Vision–Language Models (VLMs) are accurate but opaque, and concept-based explainability promises to expose their internal knowledge as human-readable units. We re-implement the MedConcept pipeline [5]—sparse autoencoder (SAE) decomposition of a frozen VLM's image embeddings, alignment to a curated radiology vocabulary, and LLM-as-judge evaluation—on BiomedCLIP [14], on *two* scales: IU X-Ray [3] (~7k images) and ROCOv2 (~80k, a 10× scale test). Our first result is methodological: the slot-wise cross-seed Jaccard used as the stability headline is *floor-by-construction*—a single SAE whose features are relabelled by a random bijection scores ≈0.0077, indistinguishable from real cross-seed—so it cannot establish non-identifiability. We therefore report a permutation-invariant *matched* metric against a subspace-conditioned (effective-rank) null. Under this honest metric, SAE concepts are genuinely non-identifiable: best-match cosine 0.30–0.33 at only 1.67–1.81× the conditioned null, with ≈0% of features matching at ≥ 0.9 across all seed pairs, and naming cosine (mean) of 0.40–0.48 (0% above 0.7) even with a chest-specific vocabulary. Crucially, the 10× ROCOv2 scale does *not* lift identifiability (matched ratio 1.67×→1.81×, frac≥0.9 0%→0.3%), refuting the data-scale hypothesis and localising the binding constraint to the *representation site*: SAE-on-CLS decomposes an already-compressed, contrastive-shaped global vector. SPLiCE [1], a deterministic decomposition over a fixed dictionary, sidesteps the estimation problem and is the sole judge-ready method; an LLM-judge scores it 81.6% Aligned (MedGemma-4B) vs. 3.3% (Llama-3.1-8B)—local plausibility orthogonal to non-identifiability.

KEYWORDS

Concept-based explainability, sparse autoencoders, medical vision–language models, interpretability evaluation, reproducibility

1 INTRODUCTION

Medical VLMs attain strong performance on pathology classification, segmentation, and report generation, yet their internal representations are high-dimensional, polysemic, and opaque—a serious concern in safety-critical clinical use that classical post-hoc tools (saliency, attention) rarely resolve into semantic structure.

Concept-based explainability reframes an explanation in terms of human-interpretable units. Where supervised approaches such as Concept Bottleneck Models [7] and TCAV [6] require a pre-defined, labelled concept set, recent work attempts *unsupervised* discovery: decomposing a frozen model's representations into sparse features, naming each feature against a clinical vocabulary, and validating the result quantitatively. MedConcept [5] instantiates this as a three-stage pipeline (sparse-activation extraction, vocabulary alignment, LLM-judge evaluation) and introduces the Aligned/Unaligned/Uncertain metrics for quantitative interpretability assessment.

A central but under-reported question is whether the concepts such pipelines discover are *reproducible* and *clinically valid*. In this work we re-implement the MedConcept paradigm faithfully, and find—rigorously, and against analytical nulls—that the learned sparse decomposition is essentially non-identifiable: independently seeded SAEs share no canonical features (a permutation-invariant matched metric lands at 1.67–1.81× a subspace-conditioned null, with ≈0% of features matching at ≥ 0.9 cosine), the naming signal is capped at 0.40–0.48 cosine, and a large fraction of features never fire. Crucially, we first expose that the slot-wise index Jaccard used as the stability headline is *floor-by-construction* (a relabelled copy of one SAE scores ≈0.0077 on it), so non-identifiability must be measured permutation-invariantly.

We probe three mitigations—a **baseline** 512-d SAE (a faithful MedConcept re-implementation, characterised as a failure case), **Path A** (a 768-d pre-projection hidden-state SAE targeting the representation-location mismatch), and **Path B** (SPLiCE, a fixed RadLex dictionary that removes the estimation problem)—plus a concept-**organisation** extension, and contribute: (1) a *relabeling control* proving the slot-wise cross-seed Jaccard is floor-by-construction (a relabelled SAE scores ≈0.0077 on it but 1.0 on a matched metric); (2) a permutation-invariant *matched-stability* metric with a subspace-conditioned null, under which SAE concepts are non-identifiable; (3) a direct refutation of the data-scale hypothesis via ROCOv2 [12] at 10× scale; and (4) an honest, partially-negative evaluation rather than a curated success story.

2 RELATED WORK

Concept-based explainability. Concept Bottleneck Models [7] restructure a network to predict an explicit concept layer before the classifier, trading minor accuracy for auditable, concept-level intervention. TCAV [6] tests directional sensitivity of a logit to user-defined Concept Activation Vectors. Both presuppose supervised, pre-defined concepts—precisely the limitation unsupervised discovery targets.

Sparse autoencoders for monosemantic decomposition. Bricken et al. [2] showed that overcomplete dictionary-learning autoencoders trained on transformer activations yield individual latents that activate on coherent, human-nameable features. Gao et al. [4] scaled the Top-K (k-sparse) autoencoder, which sets the L_0 sparsity exactly via top-k gating and removes the L1 sparsity penalty, improving the reconstruction–sparsity frontier; we adopt this architecture.

SAE stability and universality. Lan et al. [8] showed SAE feature spaces are similar only up to rotation (shared subspace, not identical features); Leask et al. [10] argued SAEs "do not find canonical units of analysis." This motivates our matched-stability metric: we expect only a shared subspace, not identical features.

Sparse concept decomposition. SPLiCE [1] bypasses a learned dictionary and decomposes a frozen CLIP embedding directly into a sparse, non-negative combination over a fixed, interpretable concept dictionary. Because the dictionary is fixed and the method deterministic, it sidesteps the non-identifiability of learned factorisation—a property we exploit as a data-frugal baseline.

Medical VLMs and the reference framework. BiomedCLIP [14] is a domain-specific contrastive VLM (ViT-B/16 + PubMedBERT, PMC-15M) whose image encoder exposes a 768-d pre-projection hidden state; its checkpoint projects this to the 512-d shared space we decompose. MedConcept [5] provides the template (sparse extraction, vocabulary alignment, LLM-judge) we re-implement; RadLex [9] the radiology lexicon for naming.

3 RESEARCH GAPS

From the literature and the project brief we identify six gaps that this work addresses:

(1) **Instability / non-identifiability:** sparse factorisations are not reproducible across seeds and provably non-identifiable in the small-data regime; no prior medical-VLM work reports seed-stability against a null.
(2) **Clinical validity:** a cosine-assigned name is cosmetic unless it tracks real pathology, yet concepts are rarely validated against ground truth.
(3) **Representation-location mismatch:** the SAE literature operates on raw hidden states, but reference pipelines fit the SAE on the already-projected, gap-bearing CLIP space.
(4) **Small-data robustness:** learned SAEs need 10^5–10^6 activations; medical corpora are 2–3 orders smaller, yielding dead features.
(5) **Flat concepts:** explanations are top-k lists ignoring anatomical/hierarchical structure.
(6) **Non-reproducible evaluation:** claims rest on single metrics whose chance floor is never reported (slot-wise index overlap is degenerate under per-seed permutation); a metric must be null-calibrated and permutation-invariant.

4 METHODOLOGY

4.1 Backbone, vocabulary, and split

We use BiomedCLIP (ViT-B/16, 768-d hidden state projected by frozen visual_projection : $\mathbb{R}^{768} \to \mathbb{R}^{512}$ into shared contrastive space; images 224×224, L2-normalised). RadLex vocabulary: text-encoded terms ranked by cosine to multi-centroid relevance (39 anchors, 13 sub-domains), top terms + 14 NIH ChestX-ray14 seeds (~1,030 terms, 512-d).

The shared 512-d space exhibits a *modality gap* [11]: a systematic offset between the image and text cones (cross-modal image–text cosine ≈0.27, centroid L2 distance ≈0.94). Following prior work we estimate it as the difference of the mean image and vocabulary embeddings and subtract it before cosine naming/decomposition.

All experiments use *two* chest-radiography corpora. IU X-Ray [3] (7,470 frontal/lateral images across 3,852 radiographic studies) is the primary, small-scale setting; ROCOv2 (~80k images) is a 10×-scale replication that lets us test the data-volume hypothesis directly. In order to avoid data leakage, we split IU X-Ray *by radiographic study* (IU key = `patient_IM-study`), so no IU study appears at the same time in both train and test sets; in opposite, ROCOv2 is simply split by image, since there is no shared patient structure to leak. The IU split yields 5,955 train / 1,515 test images (3,081 / 771 studies) with verified group overlap = 0, recomputed deterministically on every run (sorted study set, fixed seed).

4.2 Baseline: Top-K SAE on the 512-d projected space

We train a Top-K SAE [4] (ReLU($W_{\text{enc}}(x - b_{\text{dec}}) + b_{\text{enc}}$), hard top-$k$ sparsification, b_{dec} initialised to the geometric median of training data) on the 512-d image embeddings. Configuration: dictionary size D=2048, k=32, 8,000 steps, lr=5×10^{-5}, batch size 256, five seeds (0, 42, 123, 456, 789). Concept naming assigns each *live* (non-dead) decoder row to its closest RadLex term by cosine similarity, after gap correction ($W_{\text{dec}} \leftarrow W_{\text{dec}} -$ gap). Per-image pseudo-reports list the top-5 active concepts with a stated dominant concept.

4.3 Path A: Top-K SAE on the 768-d hidden state

Path A addresses Gap 3. Instead of visual_projection(·), we extract the 768-d CLS token *before* the projection and train an identical Top-K SAE on these raw (un-normalised) hidden states. Because RadLex text lives in 512-d, we bridge the two spaces with the *frozen* projection: $\hat{W}^{512}_{\text{dec}} = W^{768}_{\text{dec}} W^{\top}_{\text{proj}}$, gap-correct $\hat{W}^{512}_{\text{dec}}$, then name against RadLex as in the baseline. Dead features are detected pre-projection (a zero 768-d row projects to −gap with non-zero norm) and cross-checked with an activation-based mask.

4.4 Path B: SPLiCE on the RadLex dictionary

Path B addresses Gaps 1 and 4. We decompose each gap-corrected 512-d image embedding into a sparse, non-negative combination over the fixed RadLex text dictionary with a two-stage solver (SPLiCE-style, in place of its original Lasso): Orthogonal Matching Pursuit selects k=32 atoms, then non-negative least-squares re-solves the coefficients on that support. The dictionary is never estimated from the ~7k samples, so the method is deterministic by construction and has no cross-seed instability. The output schema mirrors the SAE explanations so the same judge can score it.

4.5 Concept-organisation extension

Addressing Gap 5, we cluster the active vocabulary into concept families by agglomerative clustering (average linkage, cosine metric). Each cluster is annotated with the most specific shared RadLex anatomical ancestor, with two degeneracy guards: leaf-root rejection and a *subtree-size canopy* that rejects candidate ancestors whose descendant count exceeds 1% of the ontology (e.g. the trivially generic "anatomical entity" node), so labels are honest rather

Table 1: Baseline Top-K SAE (D=2048, k=32) on IU X-Ray (~6k) and ROCOv2 (~80k, 10×). Matched-cosine is permutation-invariant; "sub./iso. null" ratios are observed best-match over the subspace-conditioned and isotropic nulls.

	IU X-Ray (1×)	ROCOv2 (10×)
Reconstruction cosine	0.991	0.968
Dead features (full test)	15.0–17.8%	0.5–0.8%
Matched cosine (obs)	0.299	0.327
Decoder erank	363	357
Subspace null ratio (honest)	1.67×	1.81×
Isotropic null ratio (lower bnd)	1.98×	2.16×
Frac. matched ≥ 0.9	0.0%	0.3%
Naming cosine (live, mean)	0.40	0.48
Slot-wise Jaccard (deprecated)	0.0077	0.0077

than vacuously generic. The stage emits structured per-image families plus organisation metrics (silhouette, redundancy reduction, RadLex coverage). It is standalone and does not feed the judge.

4.6 Evaluation metrics

Stability (matched, the headline). For each decoder row in seed i we take the best cosine match across all rows of seed j (decoder-cosine matching [2, 8]), a permutation-invariant statistic. The null is a *subspace-conditioned* random-vector draw: real decoder directions concentrate in a subspace of effective rank erank=$(\Sigma\sigma)^2/\Sigma\sigma^2\approx$357–363 (not the ambient 512), so we draw null vectors inside each decoder's top-erank right-singular subspace and compare against the other decoder projected into it. This controls for the data-manifold concentration that an isotropic (full-512-d) null misses; the isotropic ratio is a reported lower bound. **Stability (slot-wise, deprecated).** Cross-seed Jaccard of top-k index sets is *floor-by-construction*—not invariant to per-seed feature permutation, so a relabelled SAE scores ≈0.0077 on it (Sec. 5.2); retained only as a deprecated identical-permutation check. **Naming.** Mean cosine of the assigned term; we additionally report the fraction of features matched at ≥ 0.7 and ≥ 0.9, the concentration-robust signal (random directions cannot reach 0.9). **Faithfulness.** Point-biserial correlation of feature activations with in-distribution IU X-Ray MeSH/Problems labels, with a per-feature shuffle null. **LLM judge.** Each concept is scored Aligned/Unaligned/Uncertain against the reference report by a prompted LLM (MedGemma-4B [13] and, for cross-LLM comparison, Llama-3.1-8B); we report the raw %Aligned rate as a measure of *local* explanation plausibility against the paired report.

5 EXPERIMENTS AND ANALYSIS

5.1 The baseline is a non-identifiable failure case (Gap 1)

Table 1 reports the parity baseline on both corpora. Reconstruction is faithful (cosine 0.968–0.991) yet cross-seed feature identity is absent: matched cosine 0.30–0.33 at only 1.67–1.81× the null, ≤0.3% matched at ≥ 0.9, naming capped at 0.40–0.48 (0% above 0.7). This is genuine non-identifiability (init distinct, decoders full-rank, no collapse); the deprecated slot-wise Jaccard lands at its ≈0.0077 floor (Sec. 5.2).

Table 2: Baseline (512-d) vs. Path A (768-d), parity configuration. Matched-cosine is permutation-invariant; "obs/null" is observed best-match cosine over the isotropic null.

	Baseline 512-d	Path A 768-d
Slot-wise Jaccard	0.0084	0.0083
Analytical floor	0.0079	0.0079
Dead features (full test)	16.0%	1.7%
Naming cosine (live)	0.42	0.47
Matched cosine (obs)	0.299	0.325
Matched cosine (null)	0.151	0.124
obs/null	1.97×	2.63×
Frac. matched ≥ 0.9	≈0%	≈0%

5.2 Relabeling control: the slot-wise metric is floor-by-construction

The slot-wise Jaccard cannot distinguish "two different SAEs" from "one SAE whose columns were renamed." We verify this directly: permuting one trained ROCOv2 SAE's encoder/decoder rows by a random bijection π (function-preserving; reconstruction cosine unchanged to 10^{-4}) makes slot-wise Jaccard collapse to 0.0077 (indistinguishable from real cross-seed) while the matched metric correctly reports identity (1.0, frac≥0.9=1.0). The slot-wise signal is pure permutation noise—a binary identical-permutation check, not a stability metric. All later stability claims use the matched metric.

5.3 Neither representation depth nor data scale restores identifiability

Table 2 contrasts Path A (768-d pre-projection hidden state) with the baseline. Operating on the 768-d hidden state markedly reduces dead features (1.7% vs. 16%) and improves naming cosine (0.47 vs. 0.42). Yet the permutation-invariant matched metric is unchanged in its verdict: both methods share a decoder subspace above the null (isotropic 1.97×/2.63×, collapsing to ≈1.3–1.5× under the subspace-conditioned null), and almost no features match strongly (0% at ≥ 0.9). This is *weak universality* [8, 10]: a shared subspace exists, but not canonical, reproducible features. Crucially, Path A still operates on the *pooled CLS token* (pre-projection, but still the contrastive-shaped global summary), so the representation-site mismatch is only partially addressed.

We then test the leading alternative explanation—data scale (Gap 4)—directly. Training the parity baseline on ROCOv2 (~80k images, a 10× increase over IU X-Ray) improves training health (dead features 16%⟶0.6%) but leaves identifiability flat (Table 1): matched cosine 0.299⟶0.327, subspace ratio 1.67×⟶1.81×, frac≥0.9 0.0%⟶0.3%, naming capped at 0.40–0.48. A 10× corpus does not lift a single one of these above the non-identifiability regime. We conclude the binding constraint is the *representation site*, not data volume: SAE-on-CLS factorises a single compressed, contrastive-shaped 512-d vector whose intrinsic dimension (erank≈357–363) is far below the 2048-way dictionary, so no feature is canonical.

A dictionary/k sweep on Path A (Table 3) confirms the pattern: matched obs/null falls from 2.78× (D=1024) through 2.63× (default) to 2.00× (D=4096) as overcompleteness grows—no setting escapes weak universality.

Table 3: Path A dictionary/k sweep (768-d, 8k steps).

Preset	D	k	Dead	Matched obs/null
Conservative	1024	16	41.7%	2.78×
Default	2048	32	12.9%	2.63×
Aggressive	4096	64	6.6%	2.00×

Table 4: Concept organisation across sources.

Source	Active	Clusters	Silhouette	Empty img. (/1515)
SPLiCE	997	32	0.020	0
SAE baseline	80	9	0.094	0
SAE hidden	14	4	0.284	1260

5.4 Path B is deterministic and judge-ready

SPLiCE decomposes every test image with a fixed, randomness-free solver—Orthogonal Matching Pursuit selects the k=32-atom support and non-negative least squares re-solves the coefficients on it—so the result is deterministic by construction with no cross-seed instability, and its SAE-compatible schema lets the same judge read SPLiCE concepts. On IU X-Ray it covers all 1,515 test images in 9.5 s using 997/1,031 terms (96.7%, 14.18 concepts/image); on ROCOv2 (~80k) it covers all 15,958 images in 101 s using 1,024/1,024 terms (100%, 8.56 concepts/image). Modality-gap correction raises the maximum atom–image cosine sharply on both corpora (global maximum IU 0.49⟶0.85, ROCOv2 0.54⟶0.86), confirming the gap, not the solver, was the binding issue. The frequent terms are mixed rather than clean findings (ROCOv2 dominated by imaging modalities; the IU vocabulary has bilingual/off-topic entries)—a property of the supplied dictionaries, not of SPLiCE.

5.5 Concept organisation is dense but weakly separated

Applied to all three IU sources (Table 4), the extension behaves as designed. SPLiCE, the densest, groups 997 concepts into 32 families, cutting concepts-per-image from 14.18 to 8.09 (1.75× redundancy reduction) but with weak silhouette (0.020); the noisy IU vocabulary (Sec. 5.4) makes every cluster inherit a noisy RadLex ancestor. The rebuilt SAE baseline exposes 80 active concepts (9 families, silhouette 0.094, no empty images) that inherit the baseline's non-identifiability, and Path A's hidden source stays degenerate (14 concepts, 1,260/1,515 empty). ROCOv2 SPLiCE replicates non-degenerately (1,024 concepts, 32 families, silhouette 0.066), confirming density scales with corpus size though separation stays weak.

5.6 A small upper tail of features is genuinely faithful

Despite global instability, a faithfulness ablation (D=2048 baseline, 5 seeds) finds that 17.8% ± 0.9 of live features exceed the shuffle-null 95th percentile in point-biserial correlation with IU X-Ray MeSH/Problems labels, and 1.1% reach $|r| > 0.30$ (strongest $|r| = 0.459 \pm 0.057$, on a *mass* feature). This is a real but small upper-tail signal atop an otherwise degenerate dictionary: concepts are simultaneously *unstable cross-seed*—zero features cluster across the 5 seeds at cosine ≥0.90, and only 0.1% of live rows sit in a ≥4-seed cluster at the relaxed ≥0.70, not above a seed-tag shuffle-null (p=0.17)—and *partially faithful* (Gap 2).

5.7 The judge finds high but model-dependent local plausibility

Running the LLM-judge to convergence (temperature=0, zero parse errors) over the full test set with two backbones—the medical-tuned MedGemma-4B [13] and a general-purpose Llama-3.1-8B—SPLiCE explanations on IU X-Ray score 81.6% Aligned under MedGemma (1,230/1,508) but only 3.3% under Llama; the degenerate Path A hidden source still scores 76.3% vs. 0.5%; and the SAE baseline on ROCOv2 scores 88.3% (14,095/15,957) vs. 23.1%. Llama's dominant response is *Uncertain*. This *local* plausibility is orthogonal to the global identifiability verdict—a feature can align with one report yet fail to recur across seeds—and the cross-LLM spread shows the %Aligned rate is a property of the judging model as much as of the explanation.

6 CONCLUSIONS

We set out to discover interpretable concepts in a medical VLM without supervision, and instead produced a rigorous characterisation of where and why this fails. The slot-wise stability headline is floor-by-construction (a relabelled SAE scores ≈0.0077), so non-identifiability must be measured permutation-invariantly; under the matched metric with a subspace-conditioned null, SAE concepts are genuinely non-identifiable (best-match 1.67–1.81× the null, ≈0% at ≥ 0.9, naming 0.40–0.48). Neither Path A (768-d hidden state) nor a 10× corpus (ROCOv2) lifts it—the binding constraint is the *representation site*: SAE-on-CLS factorises a single compressed global vector. SPLiCE, on a fixed dictionary, is the sole deterministic, judge-ready alternative; a concept-organisation extension still cuts its redundancy (1.75×), and a small (~18%) upper tail of features is genuinely faithful.

Three lessons generalise: interpretability claims about discovered concepts must be calibrated against analytical nulls *and* a permutation-invariant statistic; data scale is the suspect of last resort for a failing factorisation (10× changed training health, not identifiability); and a partially-negative evaluation is more useful than a curated one.

Limitations and future work. The LLM-judge (Sec. 5.7) scores high but model-dependent local plausibility, orthogonal to identifiability; random-k null explanations score comparably (79.6% MedGemma, 2.9% Llama), indicating weak discriminability. The decisive mitigation is an SAE on the BiomedCLIP *patch-token residual stream* (mid layer, not pooled CLS) [2]—the experiment our diagnosis predicts would restore identifiability.

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